The Skew-Consensus Validation System

An Extended, Code-Backed Companion:
Agents, Workflows, and the Path from Hypothesis to Live Paper Trading

Companion to **Project Record v2.2** · expands every agent, every condition, every handoff
Twelve agents · ten-stage validation pipeline · two-agent optimisation layer · enforced paper trading
Strategy of record: **skew_consensus_v22_novix** · ~454,798 fires, 2013–2026, survivorship-free

Prepared May 2026 · branch `claude/funny-elbakyan-1e8456`
Confidential working document

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Part I — Orientation

1. How to read this document

This is the engineering companion to the one-page Project Record. Where the Record states *what* the system achieved, this document explains *how* — agent by agent, condition by condition — and every numeric threshold, rule, and handoff below is taken directly from the code (file names are cited; code itself is not reproduced).

The system answers three questions in sequence, each handled by a distinct group of agents:

1. **Is this trading idea real?** — a ten-stage validation pipeline (agents 1–10) that turns a prose thesis into a gated, statistically-defended verdict.
2. **Can we make it better without fooling ourselves?** — a two-agent optimisation layer (agents 11–12) that searches the tunable surface under hard anti-overfitting governors.
3. **Will it survive contact with a real broker?** — an enforced paper-trading loop (Phase 4) on a live options-data feed, with a live transition (Phase 5) gated on an accrued track record.

Executive summary. A prose hypothesis is formalised, adversarially screened, coded into a signal, and back-tested against a date- and direction-matched random pool. It then passes deflated-Sharpe, correlation and concentration gates, a nine-criterion qualitative review, and a risk-and-sizing officer, before being recorded and rendered into a living report. The strategy of record — a skew/IV/risk-reversal *fade* on US equities — clears every gate (21-day selection increment **+18.15 bps**, **z 8.87**), sizes to a **1.12 Sharpe / –9.86% max drawdown** book, and is deployed at **0.5×** to enforced Alpaca paper trading. The optimisation layer searched 18 variants and — correctly — promoted none: the deflated probability-of-real fell to **0.874 < 0.95** at 30 cumulative trials, so the guardrails refused to ship an overfit. The edge is real, robust, and economically modest, with one known regime weakness (a sharp V-recovery runs over the fade).

21d increment	21d gross	z-score	Sharpe (sized)	Max drawdown	Deploy size
+18.15 bps	+106.96 bps	8.87	1.12	–9.86%	0.5×
vs random pool	per 21d trade	$p < 0.0005$	CAGR +5.82%	every year +	Risk Agent 4/10

2. The big picture — three operating regimes

The system is one linear pipeline wrapped by two loops. The **linear pipeline** (stages 1→10) is the validator. The **optimisation loop** (agents 11–12) wraps it at design time: a promoted variant re-enters at stage 3→4 and clears the whole gate stack again. The **paper loop** (Phase 4) runs forward daily on a live feed once a strategy is validated and sized.

FIGURE 1 — Master pipeline: from a prose thesis to enforced paper trading

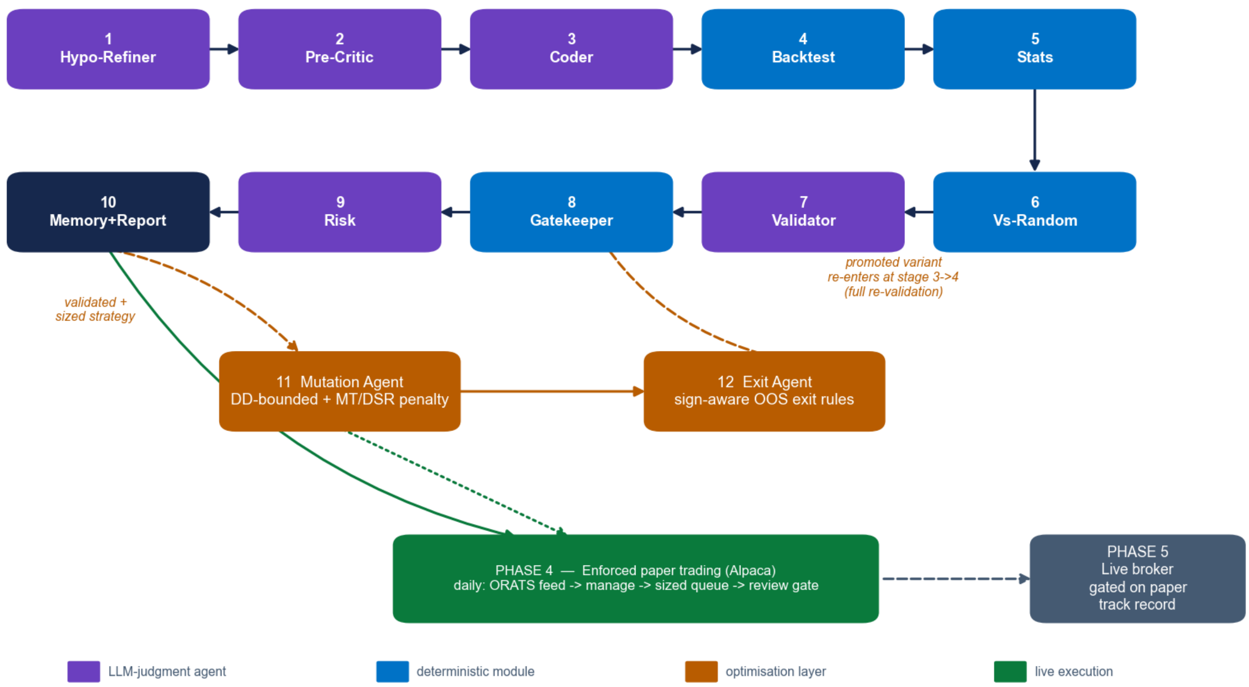


Figure 1 — The master pipeline. Purple = LLM-judgment agents; blue = deterministic modules; amber = the optimisation layer; green = live execution. Stages 1–10 validate; agents 11–12 optimise under governors; Phase 4 trades on paper; Phase 5 (live) is gated on a paper track record.

Deterministic maths, LLM judgment. The division of labour is deliberate: anything that must be reproducible and defensible (the signal, the backtest, the statistics, the gates, the sizing) is **deterministic Python**; anything requiring judgment under ambiguity (formalising a thesis, killing a bad idea, qualitative review, risk sign-off) is an **LLM agent** with a tightly-scoped rubric. Twelve agents in total: six LLM agents, the deterministic backtest/stats/gates modules, and the two optimisation agents.

3. The strategy in one page — the skew-consensus fade

The strategy reads the **ORATS options surface** to detect crowded extremes and **fades** them, trading the **underlying equity** (options inform direction only, so the traded instrument is equities). It fires when three independent readings of the surface agree:

- **M1 (corner):** the put/call percentile corner is extreme — a call-rich corner (`putP≤25 & callP≥75`) signals euphoria → short; the mirror signals capitulation → long.
- **M2 (vol + skew):** implied-vol percentile is high (`ivP≥75`) and the risk-reversal percentile is at an extreme.
- **M3 (exhaustion):** a sigma-stall (a flat, persistently-extreme move) *or* a skew-divergence (a one-sided skew that just turned) confirms the move is exhausting.

A **BULL** reading carries `signal_sign = −1` — it profits if price *falls* (the fade). A **BEAR** reading carries `+1`. The position is held 21 trading days unless an exit rule fires earlier. Filters: an earnings blackout and a short-trend filter are kept (tail control); a VIX filter was removed (it destroyed alpha).

Property	Value (code-backed)
Signal inputs	`putP, callP, ivP, rrP, sigma, skewDelta` (0–100 percentiles + two raw series)

Combination	M1 ∧ M2 ∧ M3, freshness-3, thresholds hi/lo = 75/25, sigma_threshold = 1.0
Direction	BULL → –1 (short fade); BEAR → +1 (long fade)
Hold / exit	21 trading days, with OOS-validated early exits (hard_stop 8% + Bollinger reversion band)
Universe	Survivorship-free US equities (active + delisted); ~8,519 tickers in the clean panel
Validation edge (21d)	+18.15 bps selection increment, +106.96 bps gross, z 8.87, $p < 0.0005$, beat-rate 51.6%
Sized book	Sharpe 1.12, CAGR +5.82%, max drawdown –9.86%, worst month 2020-02 –7.20%
Known weakness	A sharp V-shaped recovery runs over the fade (e.g. the 2020 rebound)

3b. Design philosophy & the anti-overfitting doctrine

The single hardest problem in systematic research is not finding a backtest that looks good — it is *not being fooled* by one. Every design choice in this system serves that goal. Two principles organise everything:

1. **Deterministic maths, LLM judgment.** Anything that must be reproducible and defensible — the signal, the backtest, the statistics, the gates, the sizing — is plain, version-controlled Python whose output is byte-stable. Anything that needs judgment under ambiguity — formalising a thesis, killing a bad idea, qualitative review, risk sign-off — is an LLM agent with a narrow, written rubric. The LLM never computes a number that ends up in a verdict; the maths never makes a judgment call.
2. **The guardrails are the product.** A loop that "keeps trying until the metric improves" is, unconstrained, an overfitting machine. The system's value is the set of mechanisms that let it search hard while refusing to deceive itself. Those mechanisms are layered, so a flaw that slips one is caught by another.

The seven defenses, each mapped to the trap it closes and the code that enforces it:

Overfitting / self-deception trap	Defense	Enforced in
"My code silently differs from the idea I validated."	Fire-parity gate — stored side must equal the reference recompute, tol 0, or scoring is refused.	`parity_gate.py` + `backtest.run`
"The market just drifted; my 'edge' is beta."	Date- & direction-matched random pool — the null is a random name on the same day, same direction.	`signal_vs_random.run_test`
"I tried 100 variants and kept the best-looking number."	Deflated Sharpe with an honest, *cumulative* trial count from the memory store.	`stats.deflated_sharpe` + `memory.n_trials`
"It only works on the data I tuned it on."	Walk-forward folds + a held-out test; the optimiser never sees the holdout during search.	`exit_agent.select_rules`, `mutation_agent._wf_score`
"A penny-stock fluke (+49900%) carries the result."	Eligibility screen: $1 price floor, |forward return| ≤ 5.0, applied exactly once.	`run_test` screen
"It's a closet duplicate of a strategy I already run."	Correlation gate vs accepted survivors (< 0.60) + PCA concentration (< 0.50).	`gates.py`
"A single bad month wipes the book."	Drawdown kill-switch (–15% → ×0.3) + fractional-Kelly caps + a hard maximum-drawdown bound on every mutation.	`sizing.py`, `mutation_agent.optimise`

> **Why "no improvement" is the headline proof.** The clearest evidence the guardrails work is that the optimisation layer — explicitly tasked with finding something better — searched 18 variants and promoted **none**, because the multiple-testing penalty correctly deflated the best candidate below the bar. A system that always "finds" an improvement is broken; this one knew when to stop.

Part II — The agent system

4. The agent roster

Twelve agents, classified by whether they exercise judgment (LLM) or compute a deterministic result. The six LLM agents are defined in `.claude/agents/*.md` (name, role, allowed tools); the deterministic stages are Python modules under `quant_validator/`.

#	Agent / stage	Type	Reads	Writes	Role in one line
1	Hypothesis-Refiner	LLM	thesis.md	refined.json	Prose → JSON spec; stress-test; never invents
2	Pre-Critic	LLM	refined.json	critique_pre.json	Adversarial kill gate, per-market templates
3	Coder	LLM	refined.json	code/signal.py	Spec → one runnable `strategy()`
4	Backtest	det.	signal.py, panel	vs_random.json + 3 csv	Fires-frame → verdict via random-pool test
5	Statistics	det.	returns.csv	metrics, dsr, walk_forward	Sharpe, DSR, walk-forward, regime, tail
6	Vs-Random	det.	annotated panel	vs_random.json	Tier-A permutation floor + Tier-B search
7	Validator	LLM	all results/*	critique_post.json	Nine-criteria qualitative review
8	Gatekeeper	det.	dsr/positions/corr	gates_outcome.json	DSR / correlation / PCA / vs-random gates
9	Risk	LLM	positions/returns/greeks	risk.json	Risk score 0–10 + size recommendation
10	Memory & Reporter	LLM+det.	decision.json	memory.db + report.html	Logs every trial; renders the living report
11	Mutation	det.	validated strategy	mutation_agent.json	DD-bounded, MT-disciplined optimiser
12	Exit	det.	fires + OHLC	exit_selection.json	Sign-aware exit-rule finder + ranker

All six LLM-agent definitions carry only `name`, `description` and an allowed-`tools` list (Read/Write/Bash/Grep as appropriate); none pins a specific model. The "Reporter" is implemented as a Skill (`report-rendering`), invoked at stage 10, not as a separate agent file.

5. Artifact data-flow & the audit trail

Every stage reads named artifacts and writes named artifacts under `theses/<id>/`. Nothing is implicit: the full chain is on disk and append-only.

FIGURE 2 — Artifact data-flow: what each stage reads and writes (theses/<id>/)

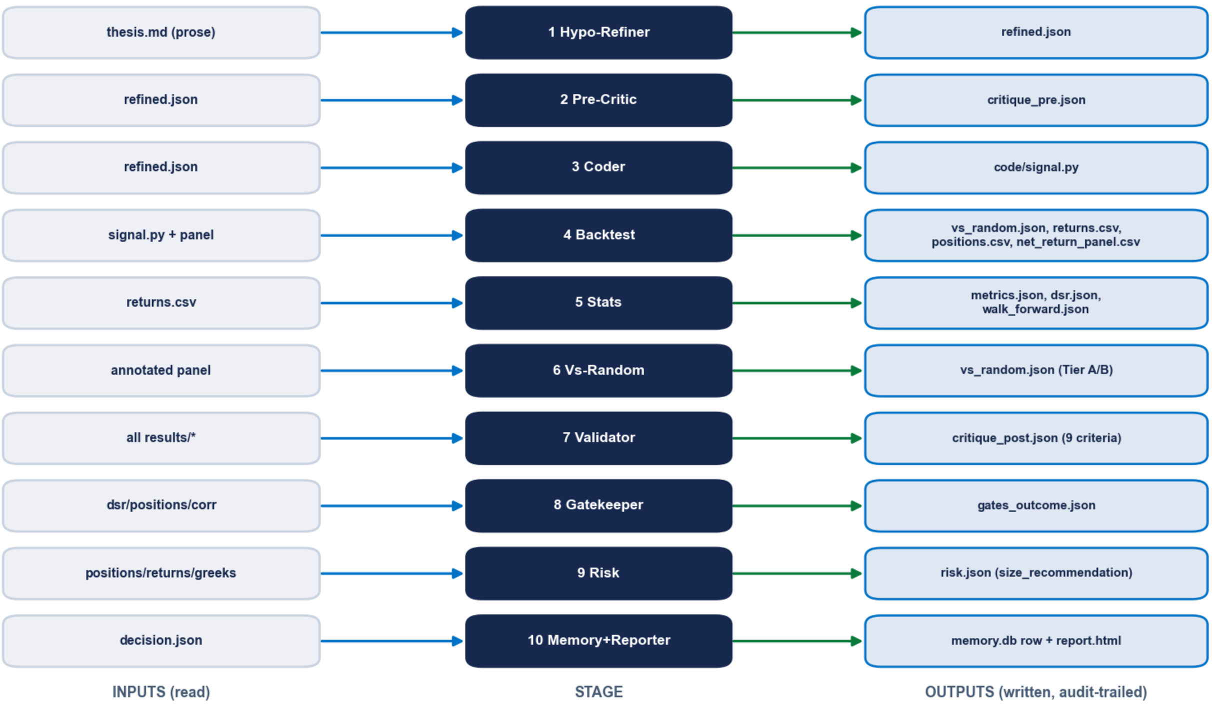


Figure 2 — The artifact data-flow. Each stage's output is the next stage's input; every write is mirrored into `audit_log.jsonl` and a human-readable `step_summaries/NN_*.md`.

The per-thesis file set: `thesis.md, refined.json, critique_pre.json, code/signal.py, results/{vs_random,metrics,walk_forward,dsr}.json + returns/positions/equity_curve.csv, critique_post.json, gates_outcome.json, risk.json, decision.json, audit_log.jsonl, user_interactions.jsonl`, and `step_summaries/01–11*.md`. Every user decision (a pause answer, an override) is logged to `user_interactions.jsonl`.

6. Mode A vs Mode B

The orchestrator auto-detects the entry mode at Step 0: if `results/positions.csv` already exists, the thesis arrives *with* results (**Mode B**, validate existing numbers); otherwise it is built from prose (**Mode A**, generate code + backtest). The skew strategy is the Mode-A regression case.

FIGURE 3 — Mode detection: A (from prose) vs B (from results)

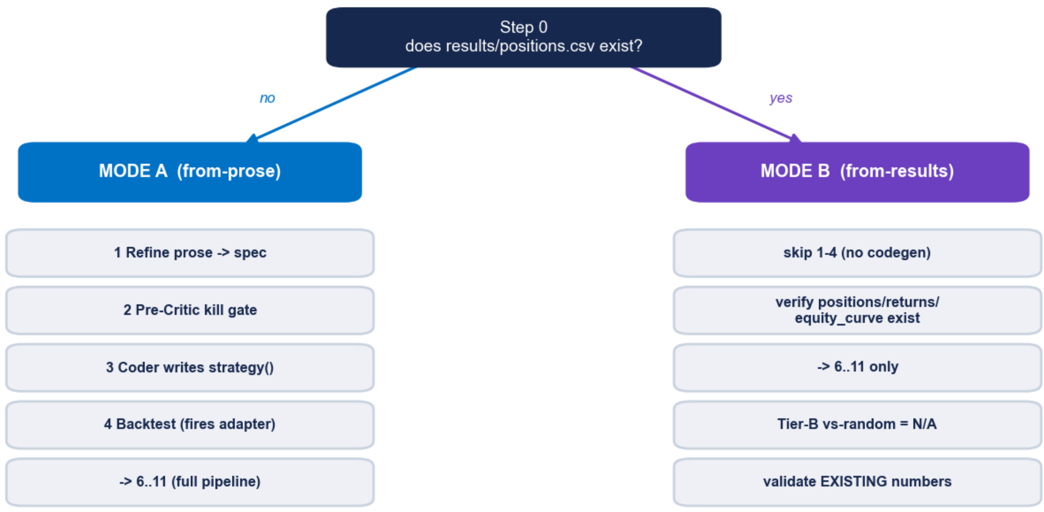


Figure 3 — Mode detection. Mode A runs the full from-prose path (refine → critique → code → backtest → …); Mode B skips code generation and validates supplied results, with Tier-B vs-random recorded as not-available.

Dimension	Mode A (from prose)	Mode B (from results)
Trigger	no `positions.csv`	`positions.csv` present
Stages 1–2	refine + pre-critic	refine + pre-critic
Stage 3–4	Coder writes `strategy()`, Backtest runs it	skipped; supplied results verified to exist
Tier-B vs-random	constraint-matched rule search runs	recorded `"not_available"`
Stages 6–11	full	full (on supplied numbers)

7. Orchestration — eleven steps, checkpoints, overrides

The `/validate-thesis` command drives the pipeline. After *every* step it prints a structured summary and appends JSON to `audit_log.jsonl`. Soft rejections halt but can be overridden with a logged justification; hard errors (sandbox/missing-data) cannot.

Step	What runs	Checkpoint / default on `--no-pause`
0	Locate/create thesis dir; detect Mode A/B	—
1	Hypothesis-Refiner	if `mode==ASK_USER`, ask single vs cross-sectional
2	Pre-Critic	on `kill` → override (`critic_pre`) or stop
3	Data fetch (Mode A)	*NotSubscribed* adapter → ask skip/abort; default **abort**

4	Coder	fire-parity fail → retry/manual/abort; default **abort**
5	Backtest (A) / verify files (B)	missing results → abort naming the file
6	Statistics (DSR with memory's n_trials)	—
6.5	Vs-Random gate	`fail` → override (`vs_random`) or stop
7	Validator (9 criteria)	`any_fatal` → override (`critic_validator`) or stop
8	Statistical gates	any fail → override (`gates:<name>`) or stop
9	Risk	`risk_score==7` → ask user; `≥8` → override (`risk`) or stop
10	Memory record	default `deployment_status="archived"` (no auto-promote)
11	Final report → `decision.json`	top line ✓ / ✗ / ■ with-override

The override mechanism (`/override-reject`). A soft rejection can be overridden only with a justification of ≥ **20 characters** (blocks "yolo"), which is appended forever to `decision.json.override_log` and the memory DB, then the pipeline resumes at *failed-step + 1*.

Overridable (soft)	Not overridable (hard)
`critic_pre, vs_random, critic_validator, gates:deflated_sharpe, gates:correlation, gates:pca, risk`	`sandbox_error, backtest_error, missing_data, missing_results_file`

If three or more overrides have already been applied to a thesis, the override command prints a standing warning before resuming — visible discipline against override-creep.

Part III — Agent deep dives

Each section states the agent's role, the artifacts it reads and writes, and the **exact conditions it applies** — every threshold, template, and rubric is taken from the agent definition or the module it drives.

8. Agent 1 — Hypothesis-Refiner

Tools: Read, Write, Grep, WebSearch. **Reads** `theses/<id>/thesis.md`; **writes** `refined.json` + `step_summaries/01_refiner.md`. It *formalises* a user's prose into a machine-checkable spec and stress-tests it — it never invents a hypothesis.

Specific rules it enforces (code-backed):

- **Mode** is one of `single` (signal per ticker independently), `cross_sectional` (ranks tickers each bar), or `ASK_USER` when genuinely unclear.
- **market_type names the TRADED instrument, not the signal source** — a strategy that uses skew/IV/VRP/GEX to trade the underlying stock is `equities`, not `options`. Allowed: `options | equities | indexes_and_index_etfs | crypto`. (This is exactly why the skew strategy is classified `equities`.)
- Both `works_in_regime` **and** `breaks_in_regime` are mandatory — the single most valuable disciplinary check.
- The stress-test must enumerate **three** specific failure modes; the warm-up buffer defaults to `max(252, 2 × lookback)` bars; `expected_sharpe_range` defaults to `[0.3, 0.8]` when the literature is unclear.
- It rejects vague theses (e.g. "oversold bounces work") demanding concrete parameters, and never modifies `user_thesis_verbatim`.

Emitted JSON keys: `hypothesis_id, title, user_thesis_verbatim, rationale, spec{signal, direction, holding_period_bars, rebalance_bars, position_bounds}, mode, market_type, expected_sharpe_range, works_in_regime, breaks_in_regime, stress_test_concerns, data_plan{adapters, universe_source, date_range, warm_up_buffer_bars}, variations`.

9. Agent 2 — Pre-Critic (four market templates)

Tools: Read, Write. **Reads** `refined.json`; **writes** `critique_pre.json`. Its mandate is blunt: “KILL bad ideas before they waste compute — when in doubt, KILL.” It routes to one of four templates by universe/signal, then applies that template’s checklist. A check tagged `[WARNING, NOT KILL]` moves the idea forward with a flag; anything else is a `kill_reason`.

Template (routing)	Checks it runs (exact)
OPTIONS (default; any IV/skew/options feature or holding options)	(1) implicit IV-surface look-ahead (ORATS EOD safe; intraday must be pinned); (2) short-vol blow-ups — require a tail hedge/stop and a backtest spanning 2008, 2018-02, 2020-03, 2022-Q1; (3) pin / early-assignment on short American legs (*long-only & European SPX/NDX exempt*); (4) costs ignoring bid–ask — mid/close fills are **FATAL**; (5) dynamic Greek hedging without a rebalance interval + costs (*buy-and-hold exempt*); (6) **[WARNING]** illiquidity — mkt-cap < $2B, front-two-expiry OI < 10,000, or ADV < $50M; (7) edge inside earnings/blackout/macro windows; (8) unnamed sector/leveraged/country ETFs.
EQUITIES	(1) factor crowding since 2010; (2) survivorship; (3) few-name concentration (TSLA/NVDA/GME); (4) earnings/OPEX edge presented as universal; (5) costs ignoring short-borrow/HTB/locate; (6) a single optimised parameter.
INDEXES & index-ETFs	(1) already-arbitraged since 2010; (2) SPX AM vs PM settlement confusion; (3) pre-2022 data extrapolated to 0DTE; (4) performance not split event vs non-event; (5) VIX/VXX/UVXY/SVXY treated as index dynamics.
CRYPTO	(1) cross-exchange timestamp look-ahead; (2) survivorship of dead tokens/exchanges; (3) funding-rate flips; (4) single-venue liquidity; (5) costs ignoring funding/gas/withdrawal; (6) leverage cascades (LUNA/3AC); (7) stablecoin de-peg untradeable.

Output: `verdict (pass|kill), market_type_applied, reasoning, kill_reasons[], warning_flags[]`. On `kill` the pipeline writes the final report and stops — overridable with `critic_pre`.

10. Agent 3 — Coder & the execution sandbox

Tools: Read, Write, Bash. **Reads** `refined.json`; **writes** one function named exactly `strategy` to `code/signal.py` (code only, no prose). The canonical output contract is the **fires frame**: one row per fire with at least `symbol, date, side∈{BULL,BEAR}, signal_sign∈{−1,+1}`, the forward returns `fwd_5/10/21`, and the stage flags `M1/M2/M3`. The Coder emits **raw** fires — the eligibility screen lives exactly once, downstream in `run_test`.

Constraint	Exact rule (House rules / SKILL.md + sandbox)
Allowed imports	**only** `numpy`, `pandas`, `math`, `ai_quant_lab.features.library`, `ai_quant_lab.features.cross_sectional`. Anything else raises `SandboxError`. (`features_custom.*` is Phase-2, deliberately not yet on the allowlist.)
No look-ahead	`.shift(1)` to make values tradeable; `.rolling(window)` for trailing stats; **never** `center=True`.
Output hygiene	positions end with `.clip(-1, 1)`; NaN treated as 0.

Legacy single-asset signatures	(A) options → `(Series, DataFrame[delta,gamma,vega,theta])`; (B) price-only → `Series∈[−1,1]`; (C) cross-sectional → MultiIndex DataFrame, weights sum to 0 or 1.

The sandbox is real. `ai_quant_lab/orchestrator/sandbox.py::run_strategy` AST-walks the source and raises `SandboxError("Disallowed import: …")` for any module outside the allowlist; it also strips dangerous builtins (`open/eval/exec/__import__`), checks the `strategy` signature, validates the returned shape/index against the input, and enforces a **10-second wall-clock timeout**. Validation is by **functional equivalence** to the reference (`compute_consensus`) via `parity_gate.assert_fire_parity` and the standing `tests/test_parity_gate.py` — the target is **0 side disagreements** on the parity tickers; there is no “sandbox validate” CLI.

11. The signal it builds — M1 ∧ M2 ∧ M3

The Coder's output reproduces `quant_validator/consensus_signal.py::compute_consensus` bit-for-bit. The signal fires only when all three readings agree on a side, within a freshness window.

FIGURE 4 — Skew-consensus signal: the M1 AND M2 AND M3 decision (per ticker, per day)

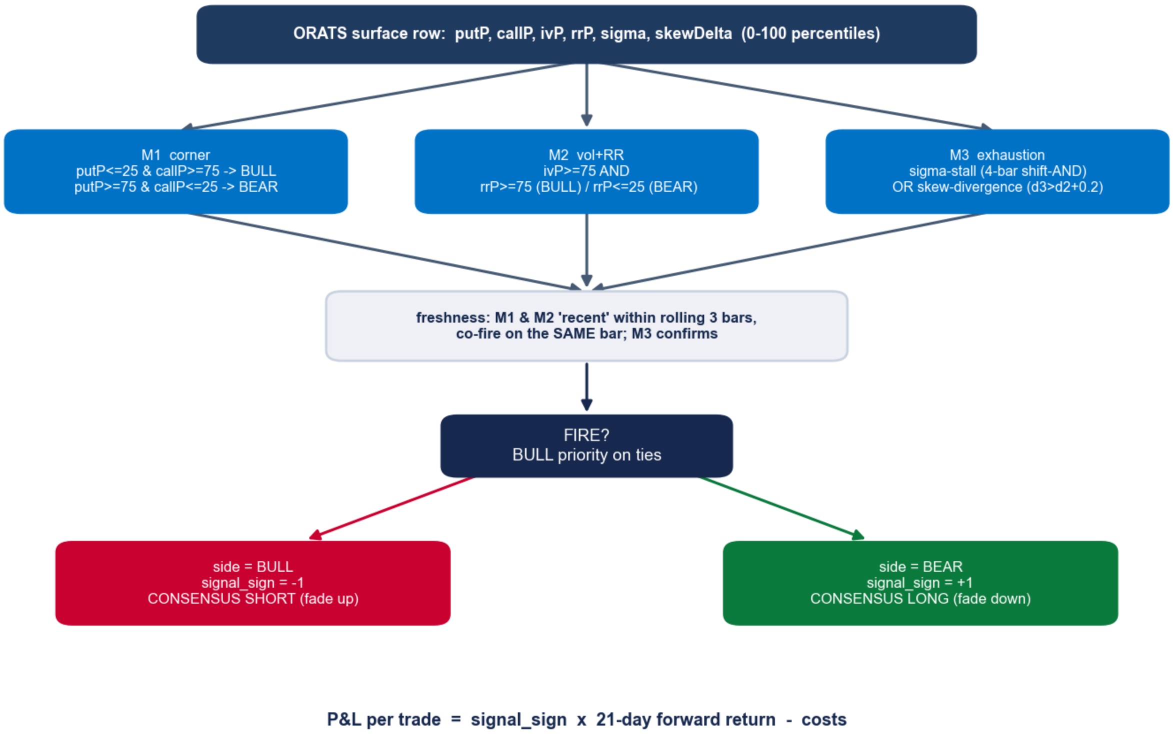


Figure 4 — The per-ticker, per-day decision. Three independent surface readings must agree; freshness lets M1 and M2 co-fire within a 3-bar window; M3 confirms exhaustion; BULL wins ties; the side maps to a signed fade.

Gate	BULL (→ short fade, sign –1)	BEAR (→ long fade, sign +1)
M1 corner	`putP ≤ 25 & callP ≥ 75`	`putP ≥ 75 & callP ≤ 25`
M2 vol + RR	`ivP ≥ 75 & rrP ≥ 75`	`ivP ≥ 75 & rrP ≤ 25`
M3 stall	`sigma.shift(3..0)` all ≥ `1.0` AND `|s3–s0| < 0.3`	all ≤ `–1.0` AND `|s3–s0| < 0.3`
M3 divergence	`d0,d1,d2 < 0 & d3 > d2 + 0.2` (skewDelta turns up)	`d0,d1,d2 > 0 & d3 < d2 – 0.2`

ConsensusOpts defaults: `hi=75, lo=25, freshness=3, sigma_threshold=1.0, sigma_mode=True, skew_mode=True, require_both=False, allow_bull=True, allow_bear=True`. M3 confirms when stall **OR** divergence holds (default; `require_both` would AND them). **Freshness** applies a trailing rolling-OR `flag.rolling(3, min_periods=1).max()` to the M1/M2 flags, and the consensus fires only where the M1 and M2 "recent" flags and the M3 confirmation are all true on the *same* bar; **BULL wins ties**. Float forms are load-bearing: `d3 > d2 + 0.2` (not `(d3–d2) > 0.2`) and a 4-bar shift-AND (not `rolling(4).sum()==4`) — algebraically equal but not bit-equal (~332 ULP cases). Economically, **P&L = signal_sign × 21-day forward return** – **costs**: a fade that profits when the crowded extreme reverts.

12. Agent 4 — Backtest (fires → verdict)

`quant_validator/backtest.py` is the keystone adapter: the single bridge from a strategy's fires frame to every downstream stage. It is generic — any strategy whose fires resolve to a `side` per (ticker, date) can be run.

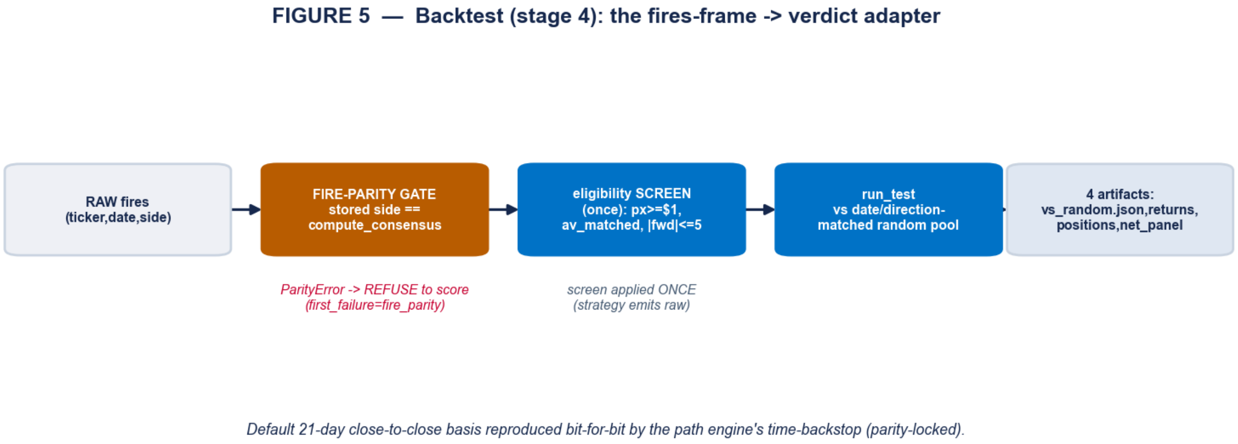


Figure 5 — The backtest adapter. The fire-parity gate runs first; the eligibility screen is applied exactly once inside `run_test`; four canonical artifacts are emitted.

- **The screen, applied once:** price floor `raw_close ≥ $1`, `av_matched` true, and `|forward return| ≤ 5.0` (caps sub-$1 penny blow-ups). Strategies emit raw fires precisely so this screen is not double-applied.
- **Four artifacts:** `vs_random.json` (the verdict), `returns.csv` (daily equal-weight book return), `positions.csv` (single-column net exposure → PCA N/A pass), `net_return_panel.csv` (per-trade panel, the sizing engine's input). Daily Sharpe is annualised with `ANN = 252`; default `cost_bps = 20`.
- **Fire-parity first:** if the declared reference is `compute_consensus`, the stored panel `side` must reproduce a fresh recompute on the parity tickers, or the stage writes `status:"fail", first_failure:"fire_parity"` and **refuses to score** (see §23).

13. The vs-random verdict engine

The verdict is not a Sharpe — it is a **selection** test. For each fire on date *t*, the engine draws random eligible tickers **on the same date** and applies the fire's own **sign**, then bootstraps (×2000). This controls for market regime *and* holds the fade direction fixed, so it measures whether the signal *picks* better than chance — not whether the market drifted.

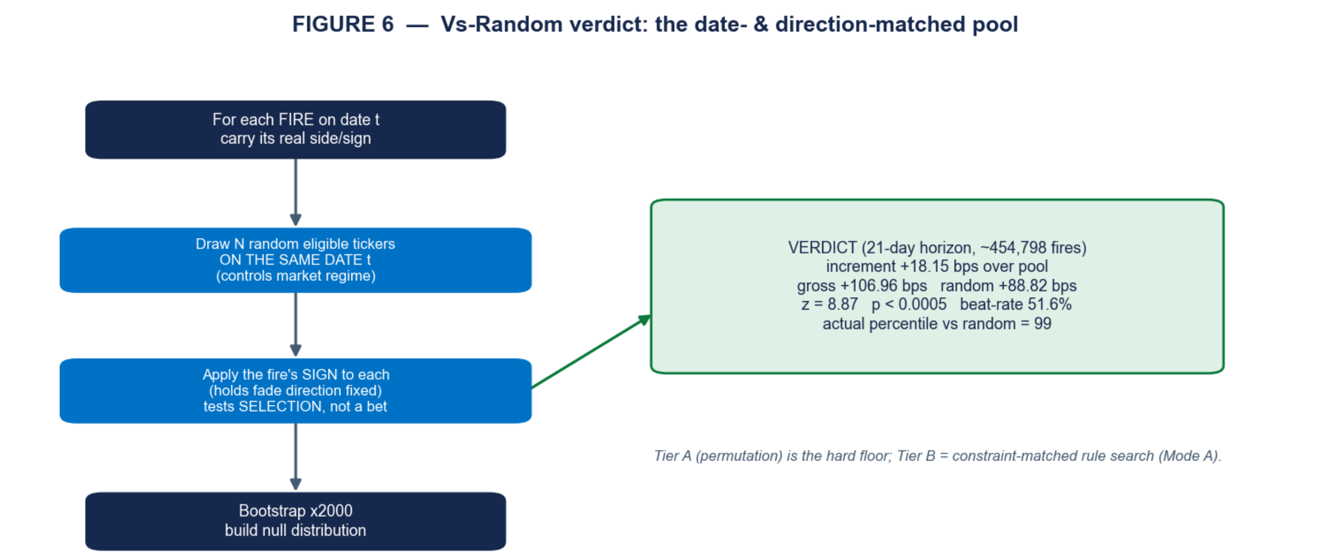


Figure 6 — The date- and direction-matched random pool. The null is "a random name on the same day, traded the same direction"; the signal must beat that null.

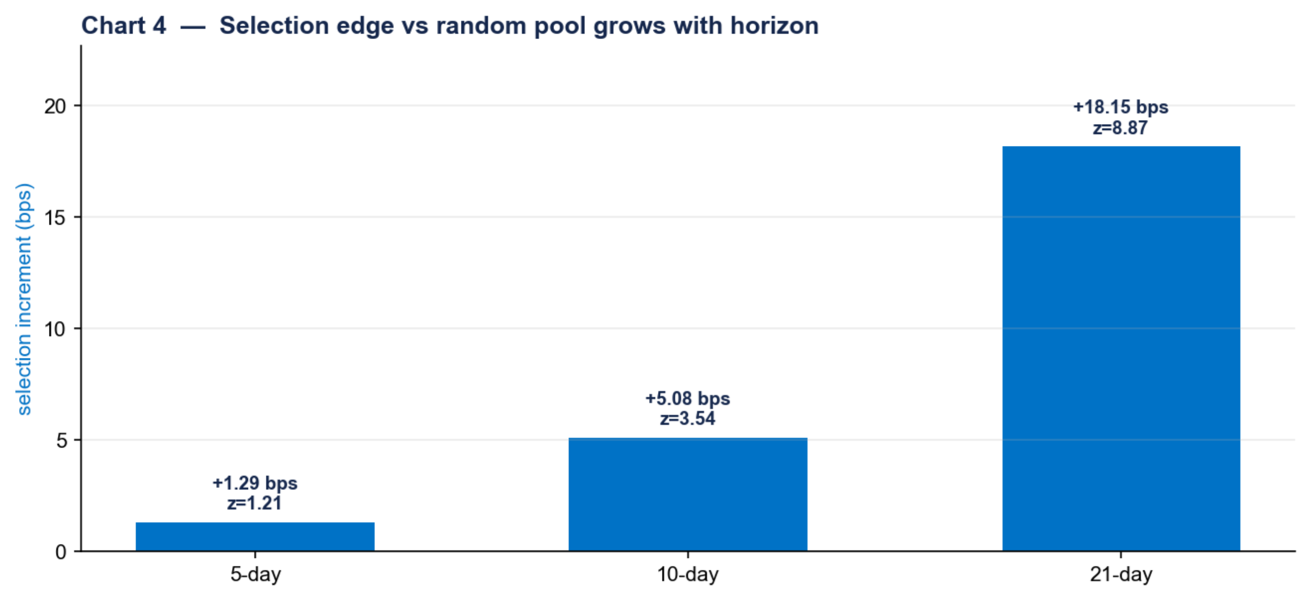


Chart 4 — The selection edge grows monotonically with horizon: +1.29 bps (5d) → +5.08 bps (10d) → +18.15 bps (21d), with z rising 1.21 → 3.54 → 8.87 (from the strategy's `vs_random.json`).

Horizon	n fires	Increment	Gross	Random	z	p	Beat-rate
5-day	456,661	+1.29 bps	14.28	12.99	1.21	0.114	50.99%
10-day	456,494	+5.08 bps	31.39	26.30	3.54	<0.0005	51.30%

21-day	**454,798**	**+18.15 bps**	**106.96**	**88.82**	**8.87**	**<0.0005**	**51.61%**

Tier-A actual Sharpe 0.98, at the 99th percentile vs the random pool. The 5-day horizon is honestly weak (p 0.11) — the edge is a 10–21 day reversion, which is exactly why the hold is 21 days.

14. Agent 5 — Statistics & the Deflated Sharpe

`quant_validator/stats.py` computes the performance metrics and — critically — the multiple-testing-aware **Deflated Sharpe Ratio** (Bailey & López de Prado).

Metric	Definition
Sharpe / Sortino / Calmar	`mean/std·√252` · downside-only denominator · `annual_return/|maxDD|`
Max drawdown + duration	min of `(equity−peak)/peak` on `cumprod(1+r)`; longest below-peak run
Distribution	skewness, excess kurtosis (Fisher), win-rate

> **Deflated Sharpe (the anti-data-mining metric).** With *N* trials and *T* observations, the expected *maximum* Sharpe under the null is `SR* = σ■_SR · [ (1−γ)·Φ■¹(1−1/N) + γ·Φ■¹(1−1/(N·e)) ]` (γ = Euler–Mascheroni), where `σ■_SR² = (1/(T−1))·(1 − skew·SR + ((kurt−1)/4)·SR²)`. The reported `dsr_probability_real = Φ((SR − SR*)·√(T−1)/denom)` is the probability the true edge exceeds that data-mined maximum (higher is better); `dsr_pvalue = 1 − that` is what the gate reads. The number of trials *N* comes honestly from the memory store, so every experiment ever run inflates the bar. For the skew strategy at validation: **p 0.006, prob-real 0.994** (n_trials = 3).

15. Agent 6 — Vs-Random gate

Stage 6.5 turns the engine of §13 into a hard floor. **Tier A** (a timing permutation that randomises *when* the strategy is in the market at the same activity rate) is the non-negotiable floor; **Tier B** (a constraint-matched random-rule search) runs in Mode A only and is recorded `"not_available"` in Mode B. A `fail` halts the pipeline (overridable as `vs_random`); a `borderline` continues with a warning.

16. Agent 7 — Validator (nine criteria)

Tools: Read, Write, Bash. It assumes the backtest is misleading and looks for the cracks, writing `critique_post.json` with a per-criterion verdict of `pass | warning | fatal`. It runs as the 4th gate, before the statistical gates.

#	Criterion	pass / warning / fatal boundary
1	Costs	spread+impact calibrated / bare fee_bps / no cost model
2	Look-ahead	`.shift(1)` clean / — / missing shift or `center=True`
3	Survivorship	point-in-time constituents / — / today's index on history (single fixed ticker auto-pass)

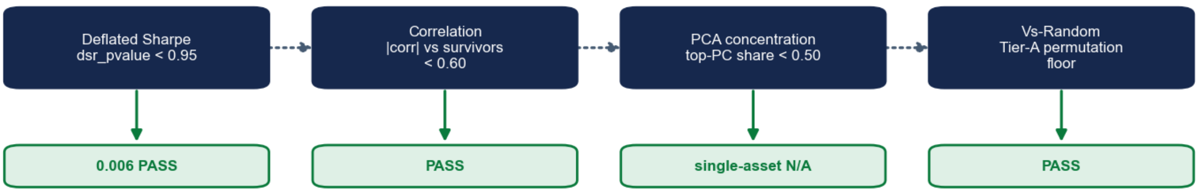
4	Optimizer constraints	caps present / missing for X-sectional / — (single-asset N/A)
5	Market impact	√-impact or <1% ADV / missing >1% ADV / large size no model
6	Participation cap	`max_pct_adv ≤ 10%` / missing at size / ignores ADV on illiquid
7	Deflated Sharpe	`dsr_pvalue < 0.95` / `0.85–0.95` / `≥ 0.95`
8	Regime-conditional	Sharpe + in all regimes / one regime only / negative in any (folds in the vs-random Tier-A verdict)
9	OOS holdout	true untouched holdout / walk-forward alone / no OOS

Output: `any_fatal`, `warnings_count`, and a `recommendation∈{reject, accept_with_warnings, accept}`. Bias: when in doubt → warning. Any fatal halts (overridable as `critic_validator`).

17. Agent 8 — Gatekeeper (statistical gate stack)

`quant_validator/gates.py` applies four deterministic gates with hard default thresholds.

FIGURE 7 — Stage 8 statistical gate stack (soft-overridable on fail)



Any FAIL halts the pipeline but is overridable via /override-reject (>=20-char justification, logged forever).
Hard errors (sandbox/missing data) are NOT overridable. The Mutation Agent reuses this exact stack per variant.

Figure 7 — The stage-8 gate stack and the strategy's result at each gate. The Mutation Agent reuses this exact stack per variant.

Gate	Pass condition	Edge behaviour	Skew result
Deflated Sharpe	`dsr_pvalue < 0.95`	N/A if dsr.json missing	0.006 — PASS
Correlation	`max|corr| with survivors < 0.60`	**pass** if no survivors	PASS
PCA concentration	top-PC variance share `< 0.50`	single-asset → N/A pass	N/A pass
Vs-Random	Tier-A permutation floor (maps stage 6.5)	N/A if file missing	PASS

The roll-up is `fail` if any gate fails, else `warning`, else `not_available`, else `pass`. A `fail` is a soft-overridable halt; a missing upstream artifact is explicitly *not* a fail. (Both `stats.py` and `gates.py` note that the 0.95 DSR ceiling is deliberately lenient for research and should be tightened toward 0.05–0.10 for production.)

18. Agent 9 — Risk & the sizing engine

Tools: Read, Write, Bash. The Risk agent runs `quant_validator/risk_stats.py` to compute deterministic statistics, scores risk 0–10, and recommends a deployment size as a fraction of the approved size; the `quant_validator/sizing.py` engine then turns approved trades into a sized book.

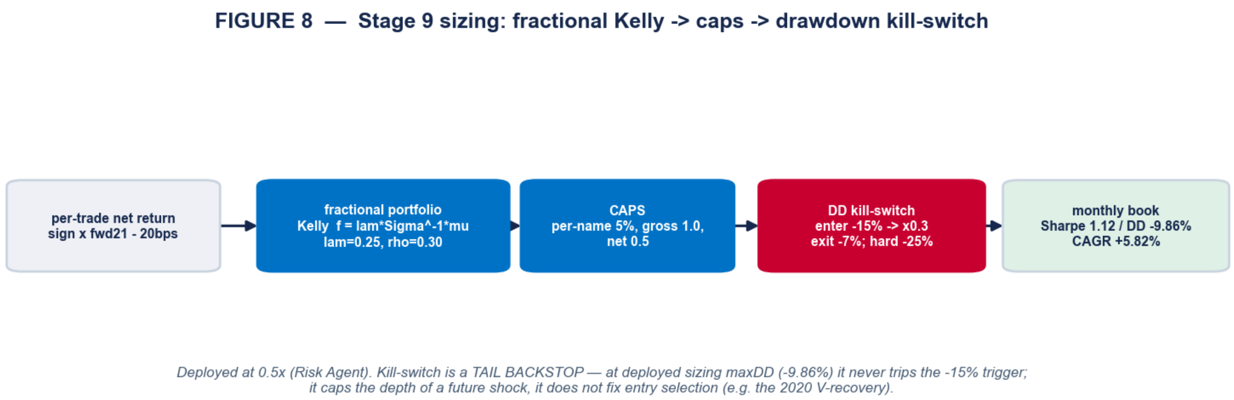


Figure 8 — The sizing pipeline: per-trade net return → fractional portfolio Kelly → caps → drawdown kill-switch → monthly book.

Risk-score lever	Trigger
Cap size to 0.4 / 0.6	net short vega without / with a documented tail hedge
Cap size to 0.5	excess kurtosis > 6, or > 50% of positions inside an event window (unless event-targeted)
Warnings (no cap)	kurtosis > 3, concentration share > 0.4, max-ticker > 25%, worst-1d –10…–15%, DD duration > 90 days
Hard reject (score ≥ 8)	regime asymmetry + net short vega, concentration > 0.6, worst-1d < –15% (non-deleveraging), max-ticker > 35%
Borderline (score = 7)	pause and ask the user (cap 0.3 / override to 0.5 / reject)

The sizing engine (deployed constants): per-trade `net_return = signal_sign · fwd_21 − cost_bps/10■` (cost 20 bps); fractional portfolio Kelly `f = λ·Σ■¹·μ` with `λ = 0.25` on a constant-correlation covariance (`ρ = 0.30`); caps applied in order — per-name **5%**, gross **1.0**, net **0.5**; and a hysteretic **drawdown kill-switch** that de-risks to **×0.3** when realised drawdown hits –**15%** and restores at –**7%** (hard flatten at –25%). Trades are bucketed into monthly cohorts (21-day hold ≈ one month). Result: **Sharpe 1.12, CAGR +5.82%, max drawdown** –**9.86%**, every year positive; the Risk agent scored 4/10 and recommended **0.5×**.

> **Honest note on the kill-switch.** At deployed sizing the book's worst drawdown (–9.86%) never reaches the –15% trigger, so the kill-switch does not engage in the deployed configuration — it is a *tail backstop* that bounds the depth of a future shock, not an entry filter. It cannot fix a selection miss such as the 2020 V-recovery; only the entry signal and the exits address that.

19. Agent 10 — Memory & Reporter

Memory (`quant_validator/memory.py`) is an append-only SQLite store. The `trials` table records every accept/reject with the honest DSR trial count at the time (`n_trials_at_time`), the metrics, and an `override_log_json`; a `trial_greeks` side table tracks net delta/gamma/vega for portfolio-level Greek aggregation. `deployment_status ∈ {archived(default), paper, live, retired}`; `n_trials()` = `COUNT(*)` feeds the Deflated Sharpe, and retired predecessors are excluded from the correlation gate so a successor isn't blocked by the version it replaces. Overrides are never lost (a placeholder row is inserted if needed).

Reporter (the `report-rendering` Skill, stage 10) maintains **one living, self-contained HTML report per strategy**: a canonical `report.json` (each stage writes only its own slice — append-and-update, never wipe) and a `report.html` that is a pure function of it, with charts baked as **inline SVG** (no CDN/JS, prints to PDF). Sections render in order: stages 1–10, a robustness block (cost-survival, temporal, regime), a phases block (Phase-4 paper / Phase-5 live with the owned-positions table, the ranked exit options, and the next-open trade queue), and a Mutations section. On a full re-run the version bumps and the prior HTML is snapshotted.

Part IV — The optimisation layer

Stages 1–10 answer “is it real and how big?” The optimisation layer answers “can we make it better without fooling ourselves?” Its primary job is to **search hard while refusing to overfit** — the guardrails are the product.

20. Agent 11 — Mutation (the constrained optimiser)

`quant_validator/mutation_agent.py` is a closed loop: propose a guarded variant → re-fire and path-backtest it across walk-forward folds → risk-score it against the baseline and the drawdown bound → feed back → propose again. It is **thesis-locked** (the M1∧M2∧M3 edge is never mutated, only knobs around it) and bounded by two governors.

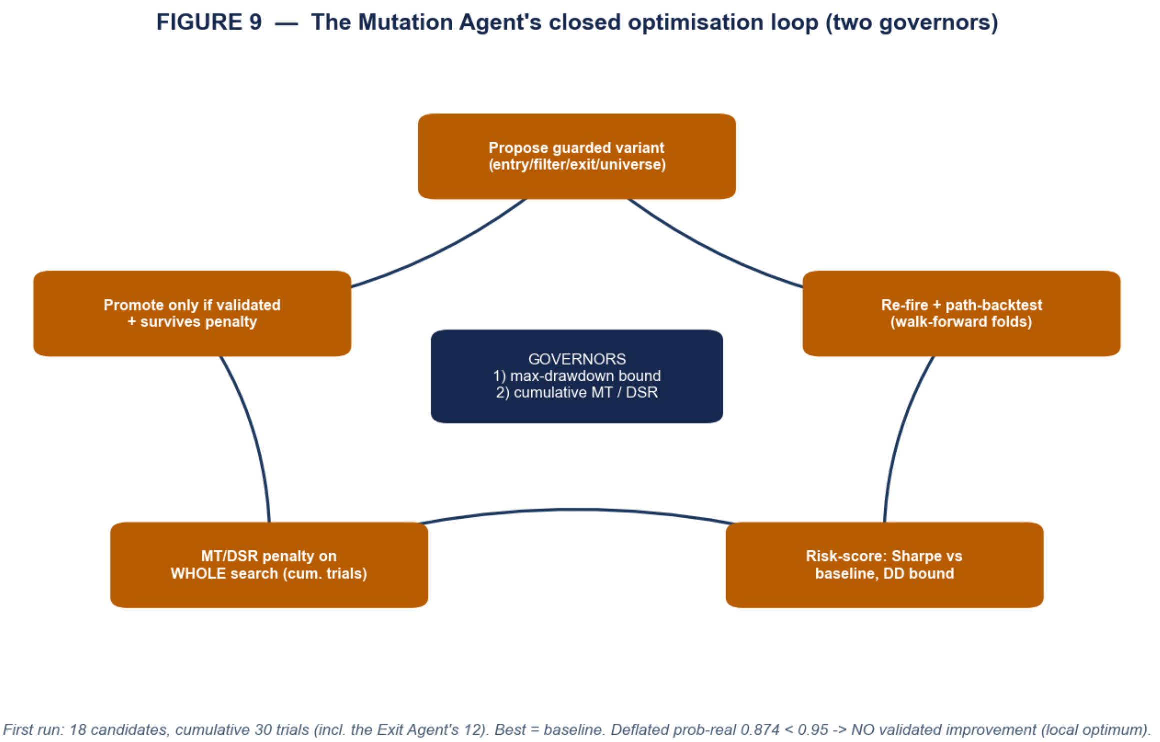


Figure 9 — The closed optimisation loop and its two governors: a hard maximum-drawdown bound, and a multiple-testing / Deflated-Sharpe penalty applied to the whole search.

Candidate set	Members (code-backed)
Exit (11, cheap — reuse fires + OHLC)	hard-stop only at 8/6/10/5%, Bollinger band only, hard-stop+Bollinger mean, trailing+Bollinger, base+vol-spike, base+squeeze, base+tail, hard-stop+profit-target
Universe (2, cheap — fire filter)	liquid `raw_close ≥ 5`, liquid ≥ `10`
Entry (4, expensive — re-fire consensus)	corner 70/30, corner 80/20, freshness 5, sigma 1.5

Promotion is hard. A variant must beat the baseline Sharpe on a **majority** of folds, keep `pooled maxDD ≥ −0.25`, retain ≥ **50%** of the baseline mean edge, and beat the pooled Sharpe — *and then* survive the multiple-testing penalty. The Deflated-Sharpe trial count is **cumulative**: `n_trials = 12 (the Exit Agent’s prior candidates) + 18 (this run) = 30`, so every variable tried inflates the bar for the best one.

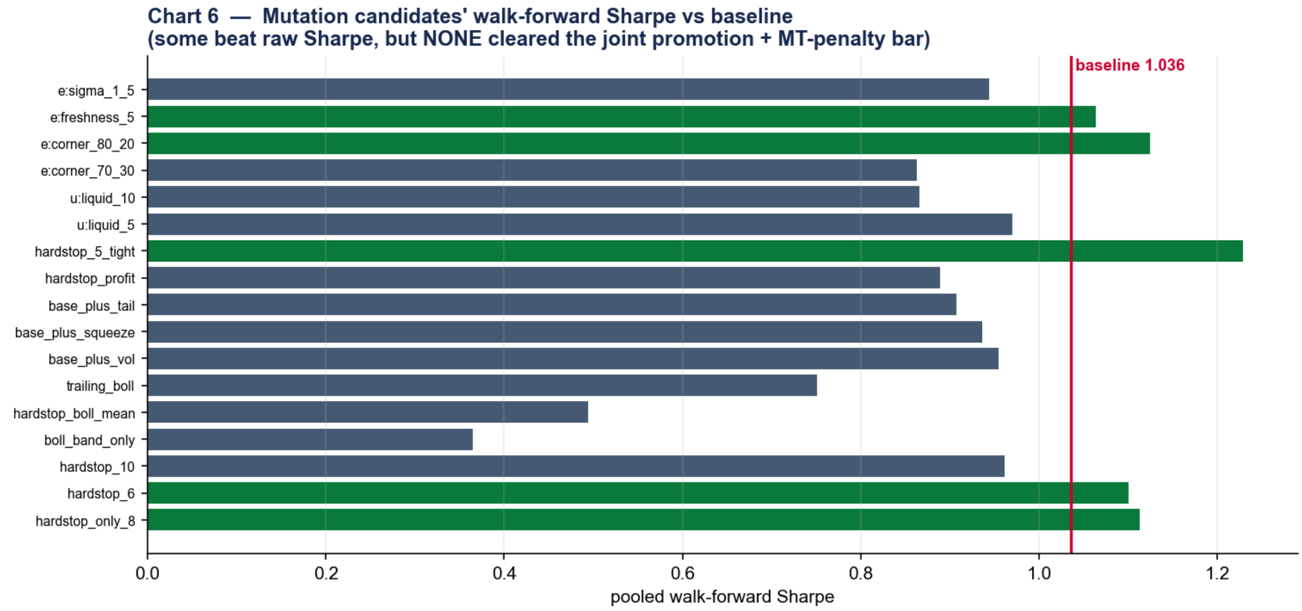


Chart 6 — Walk-forward Sharpe of all 18 variants vs the baseline (red line, 1.036). Several beat raw Sharpe — `exit_hardstop_5_tight` 1.229, `entry_corner_80_20` 1.124 — but none cleared the joint promotion + multiple-testing bar.

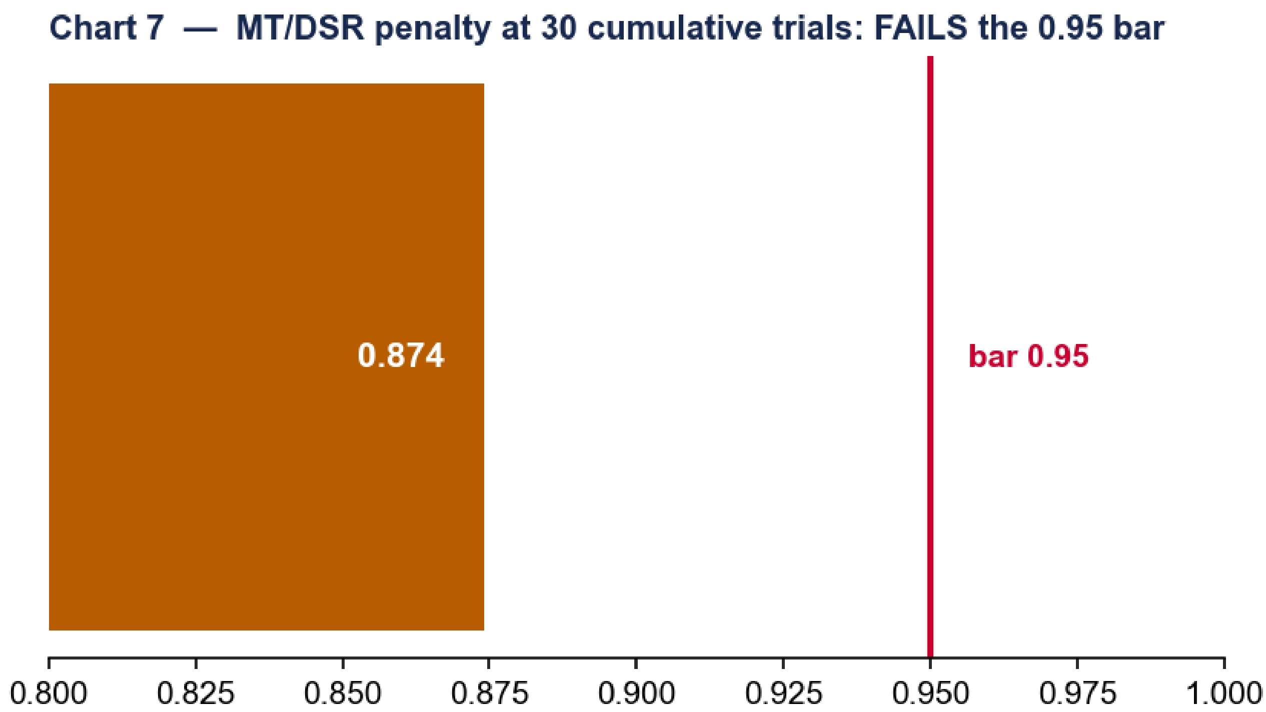


*Chart 7 — The decisive test. At 30 cumulative trials the best variant's deflated probability-of-real is 0.874, below the 0.95 bar. The optimiser correctly promoted **nothing**.*

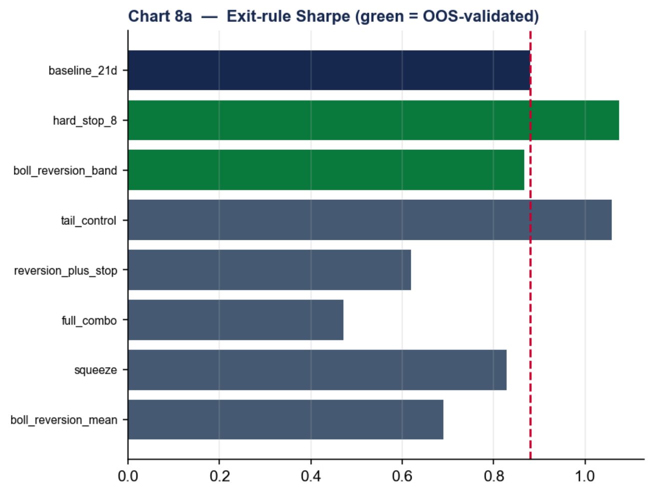
> **Result: NO VALIDATED IMPROVEMENT — and that is a success.** 18 candidates, 30 cumulative trials, runtime 88s. Best variant = the baseline (v22 + 6b exits): Sharpe 1.036, maxDD –5.25%. The expected maximum Sharpe under the null at 30 trials is 0.036; the best annualised Sharpe was 0.891 → deflated prob-real 0.874 < 0.95. The guardrails refused to ship an overfit; the strategy sits at a local optimum.

21. Agent 12 — Exit (technical-analysis exits)

`quant_validator/exit_agent.py` replaces "always ride to day 21" with finding the exit that captures the most profit before the fade reverts. It has three parts: a sign-aware rule library, an out-of-sample walk-forward selection, and a run-time ranker.

Rule builder	Default params	Purpose
`r_time`	21 bdays	non-negotiable time backstop
`r_hard_stop`	8%	tail / loss control
`r_trailing_stop`	10%	give-back control
`r_profit_target`	15%	lock a win
`r_bollinger_reversion`	n=20, k=2.0, target mean|band	the fade's take-profit (entry-context-aware)
`r_vol_spike`	yz5 or atm_iv ≥ 0.80	regime-turn exit
`r_squeeze`	skew ≤ –2.0	short-side squeeze tail control (the 2021-01 answer)

OOS selection (20,000-fire sample, 5 time folds, 20 bps): a rule survives only if it beats the baseline Sharpe on a majority of folds *or* cuts max drawdown on a majority, **and** keeps ≥ 50% of the baseline mean edge. The baseline (21-day backstop) scored Sharpe 0.879, maxDD –23.95%, –608.5 bps on 2021-01.



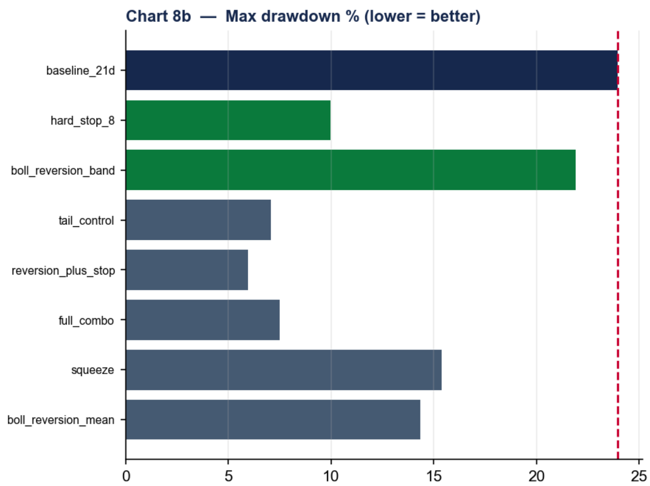


Chart 8 — Exit-rule Sharpe (left) and max drawdown (right). Green = OOS-validated. `hard_stop_8` lifts Sharpe to 1.08 and cuts drawdown to –10%; `boll_reversion_band` keeps ~97 bps of edge while cutting drawdown on all five folds.

Rule	Survives?	Sharpe	Max DD	Mean bps	2021-01 vs base
baseline 21d	—	0.879	–23.95%	100.7	—
hard_stop_8	**YES**	**1.075**	**–9.97%**	58.9	**+516.2 bps**
boll_reversion_band	**YES**	0.867	–21.9%	**96.7**	+63.1 bps
tail_control	no (edge-keep)	1.059	–7.06%	42.8	+675.3 bps
vol_spike_atm_iv	no	–0.528	–14.48%	–13.7	+502.8 bps

Deployed policy: `hard_stop_8 + boll_reversion_band + 21-day backstop` (sign-aware). At run time the ranker surfaces 2–3 exits per open position by urgency (`act_now > target > backstop > watch`). The protective rules are enforced in the live loop; the profit-target is advisory by default.

Part V — Backtest fidelity

22. The path-dependent engine

`quant_validator/backtest_path.py` walks daily adjusted OHLC bar-by-bar so the exit rules can see intrabar touches the close-to-close panel cannot. Rules are evaluated in priority order; the **first** rule to trigger on the **earliest** bar wins.

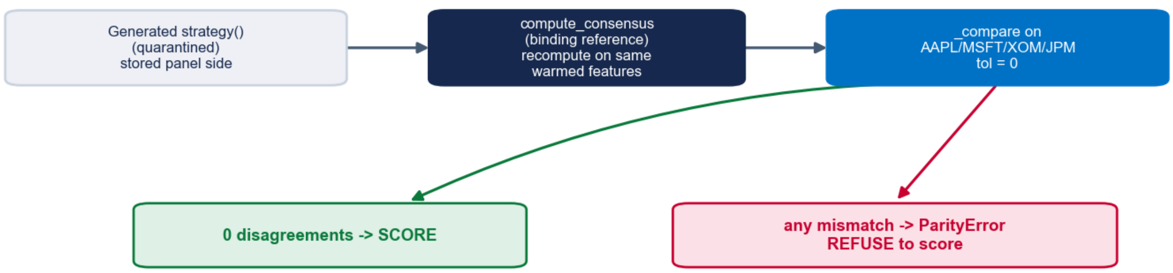
Rule	Long / short trigger (sign-mirrored)
hard_stop / trailing_stop	low ≤ entry·(1–p) / high ≥ entry·(1+p); trailing tracks run-max/run-min
profit_target	high ≥ entry·(1+p) / low ≤ entry·(1–p)
bollinger_mean / band	price reverts to the SMA-n mid / to entry ± k·σ band
bollinger_reversion	entry-context-aware: arms only if entered on the extended side, then targets mean or band
vol_spike / squeeze	Yang-Zhang 5-day or ATM-IV ≥ threshold / (short only) skew ≤ –threshold
time_backstop	t ≥ entry + 21 bdays — the non-negotiable floor

The default `entry_mode = "signal_close"` fills at the signal-date close, preserving the close-to-close basis the verdict was validated on (the alternative `"next_open"` mirrors live OPG fills). Crucially, when the only rule is the time backstop, the engine reproduces `av_fwd_21_total` bit-for-bit — a fast path the parity gate verifies to `< 1e-9`.

23. Parity discipline

A generated strategy must not reach scoring if it disagrees with its declared reference. `quant_validator/parity_gate.py` enforces this.

FIGURE 10 — The standing Coder->Backtest fire-parity gate



Runtime gate enforces MATERIALIZATION parity; an injected BULL<->BEAR swap MUST raise ParityError (3 standing tests).

Figure 10 — The fire-parity gate. The runtime gate enforces materialization parity (the stored side that `run_test` actually scores == a fresh `compute_consensus` recompute); generated-vs-reference is held as a standing test.

- Comparison runs on a fixed cross-sector ticker set `AAPL, MSFT, XOM, JPM` over the six feature columns, with tolerance **0** — any single disagreement raises `ParityError` and the backtest writes `first_failure:"fire_parity"` and refuses to score.
- Three standing tests guarantee the gate bites — including one that injects a **BULL↔BEAR swap** and asserts a `ParityError` is raised. A novel thesis that declares no reference skips the gate cleanly.

23b. The backtest-realism ladder

A backtest result means little without knowing how realistic the simulation was. This system climbs a deliberate ladder of fidelity, and — crucially — proves each rung reproduces the one below it, so added realism never quietly changes the verdict.

Rung	What it adds	Realism caveat it removes	Where
1 — Selection verdict	Signed forward returns vs a same-day, same-direction random pool	"It's just market drift / beta."	`run_test`
2 — Cost-aware	20 bps round-trip charged on every trade	"Frictionless fills flatter the edge."	`cost_bps` in adapter + sizing
3 — Eligibility-screened	$1 floor, |fwd| ≤ 5.0, AV-matched, survivorship-free universe	"Penny flukes / dead tickers carry it."	`run_test` screen + clean panel
4 — Path-dependent	Bar-by-bar adjusted OHLC so intrabar stop/target/Bollinger touches are seen	"Close-to-close hides the real exit."	`backtest_path._simulate`
5 — Parity-locked	Time-backstop path reproduces close-to-close to < 1e-9; live `_simulate` == backtest, rule-for-rule	"The live engine differs from the test."	`parity_gate`, `exit_parity_check`
6 — Live OPG	Market-on-open fills, shortability gate, idempotent orders, staleness guard	"It can't actually be traded."	`fills.py` (Alpaca)

The default `entry_mode = "signal_close"` keeps rungs 1–5 on the exact close-to-close basis the verdict was validated on; switching to `"next_open"` moves to the live OPG basis on purpose. Because each rung is parity-checked against the previous one, the +18.15 bps / z 8.87 selection verdict and the 1.12-Sharpe sized book are the *same* result viewed at increasing realism — not three different backtests.

Part VI — From idea to live

24. The data layer

The signal is only as good as the panel under it. The data layer is survivorship-free and point-in-time by construction.

Component	What it is
ORATS surface	Full-universe by-date backfill (~5,805 tickers/day); the six signal features recomputed bit-identical to the reference tool (252-day rolling percentiles; sigma = 5-day move / 20-day vol)
Survivorship	AV `LISTING_STATUS` builds an active + delisted master; `symbol_map` resolves ORATS→AV (exact then punctuation-normalised; unmatched never silently dropped)
Path bars	AV `TIME_SERIES_DAILY_ADJUSTED` (premium) → split/dividend-adjusted OHLC, the path engine's input
Clean panel	`signal_panel_clean.parquet` — merged signal × adjusted OHLC, ~8,519 tickers, pre-computed forward returns; 454,798 fires, 2013–2026
Warm-up	`WARMUP_BDAYS = 756` (3yr ORATS), `WARMUP_START = 2013-11-26` — every percentile/sigma window is fully warmed
Live feed	`run_live` backfills the gap, rebuilds the trailing-756 panel, fires the latest session (~150–250/day), and flips `is_live_orats` — it verifies, it does not trade

25. Paper-trading execution

`quant_validator/fills.py` enforces both entry and exit on Alpaca through a pluggable `FillSource` seam (`modeled | alpaca_paper | alpaca_live`). Phase 5 (live) is the same code with `paper=False`, gated on an accrued track record.

FIGURE 11 — Phase-4 daily paper loop (Alpaca, DRY-RUN by default)

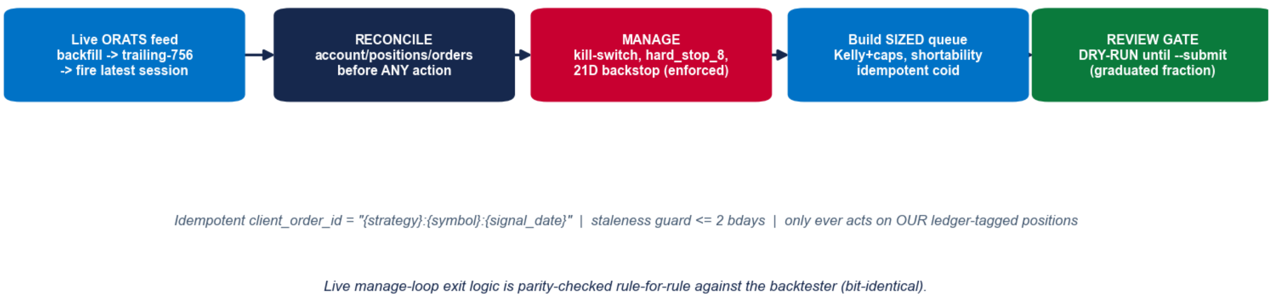


Figure 11 — The Phase-4 daily loop. DRY-RUN by default; nothing fires without an explicit, graduated `--submit`.

Safety default	Rule (non-negotiable)
Idempotency	`client_order_id = "{strategy}:{symbol}:{signal_date}"` — Alpaca rejects duplicates, so a re-run is a no-op
Reconcile-first	account / positions / open-orders are reconciled before **every** action
Only act on ours	the loop touches only ledger-tagged positions — never an unknown position
Staleness guard	refuse a signal older than **2 business days** (unless `--allow-stale`)
Caps	per-name 5%, gross 1.0, net 0.5 — a breach hard-aborts the submit
Shortability	non-shortable shorts are skipped + logged, not hard-rejected
Order type	market-on-open (`OPG` TIF), mirroring the backtest's next-open basis
Exit enforcement	`ENFORCE_EXITS = risk_only` (default): kill-switch + `hard_stop_8` + 21D backstop enforced; profit-target advisory. `enforced` adds the Bollinger reversion band.

The manage loop reconciles, runs the kill-switch (hard-flatten at –25%), enforces the protective exits via the same `_simulate` used in the backtest (parity by construction — `exit_parity_check` must be 100% bit-identical), and runs the scheduled 21-day closes. Sessions are US-holiday-aware. **Current live book (DRY-RUN):** a queue of 194 OPG orders (110 long / 84 short; 21 intended shorts dropped non-shortable) for the 2026-05-26 open, gross 0.4337 / net 0.0898 at 0.5×, held behind the first-day review gate.

26. End-to-end walkthrough — a worked lifecycle

This is the payoff: how a strategy an analyst *incubates* as a sentence travels the whole system to a paper fill.

FIGURE 12 — End-to-end lifecycle: a user-incubated thesis to a live paper fill

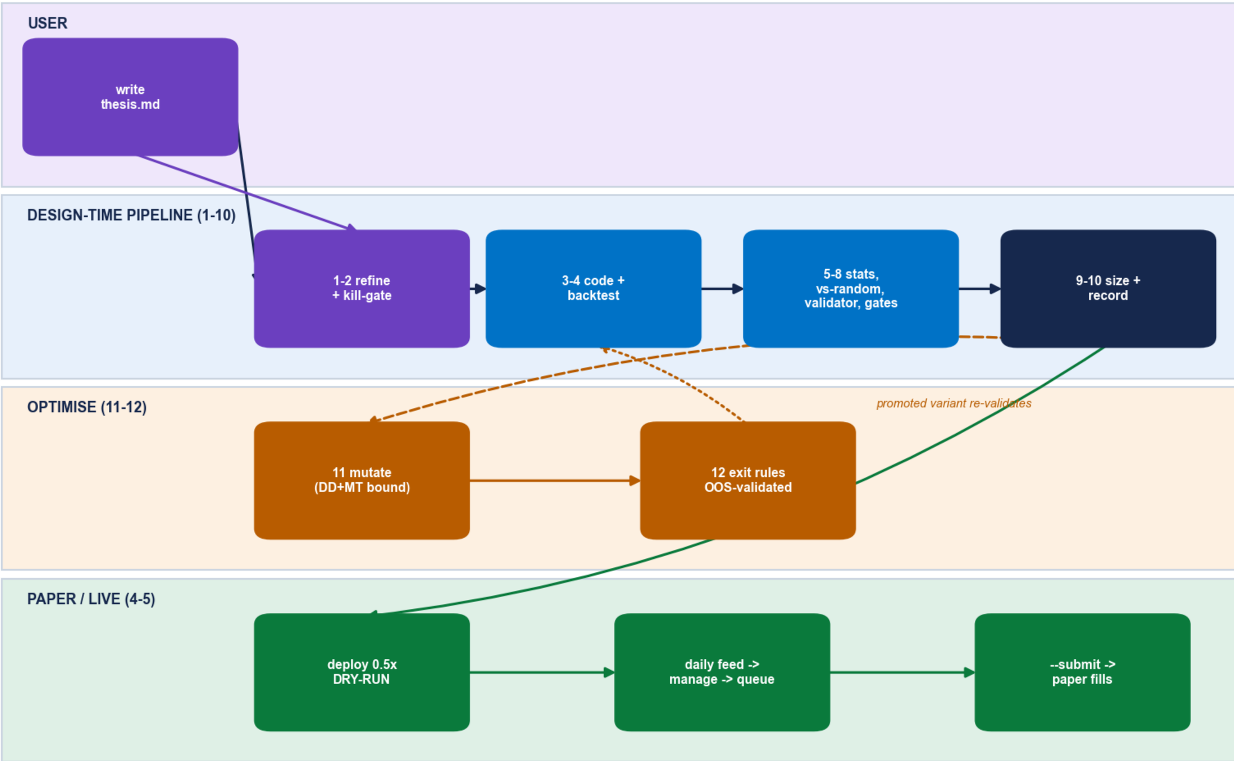


Figure 12 — The full lifecycle across four swimlanes: the user, the design-time pipeline (1–10), the optimisation layer (11–12), and paper/live execution (4–5).

1. **Incubate (user).** The analyst writes a prose thesis into `thesis.md` — e.g. “when the options surface shows a crowded, exhausted skew extreme, fade the underlying over ~a month.”
2. **Refine (1).** The Hypothesis-Refiner formalises it: `market_type = equities` (it trades the stock), `mode = single`, holding 21 bars, with a mandatory `works_in_regime` / `breaks_in_regime` pair and three named failure modes.
3. **Screen (2).** The Pre-Critic routes to the OPTIONS template and runs its eight checks (look-ahead, short-vol tails, costs, illiquidity, event windows…). Survivable concerns become `warning_flags`; a fatal flaw would kill it here.
4. **Code & gate (3→4).** The Coder writes the fires-frame `strategy()`; the fire-parity gate confirms it reproduces `compute_consensus` on AAPL/MSFT/XOM/JPM with 0 disagreements, then the Backtest applies the eligibility screen once and scores it.
5. **Prove the edge (5→8).** Statistics report a Deflated-Sharpe p of 0.006 (prob-real 0.994); the Vs-Random gate confirms +18.15 bps / z 8.87 at 21 days; the Validator’s nine criteria pass; the Gatekeeper’s DSR / correlation / PCA / vs-random gates all clear.
6. **Size & record (9→10).** The Risk agent scores 4/10 and recommends 0.5×; the sizing engine produces a Sharpe-1.12 / –9.86%-drawdown book; Memory logs the trial and the Reporter renders the living HTML report.
7. **Optimise (11→12).** The Mutation Agent searches 18 variants under the drawdown + multiple-testing governors and promotes none (0.874 < 0.95); the Exit Agent validates `hard_stop_8 + boll_reversion_band` out-of-sample and they are banked for risk control.
8. **Paper (4) & live (5).** The strategy deploys at 0.5× in DRY-RUN; each day the live ORATS feed → manage loop (enforced exits + kill-switch) → sized, shortability-gated queue → first-day review gate. An explicit, graduated `--submit` begins real paper fills; Phase 5 (live broker) is the same code, gated on the accrued paper track record.

Part VII — Results & appendices

27. Results gallery & the honest read

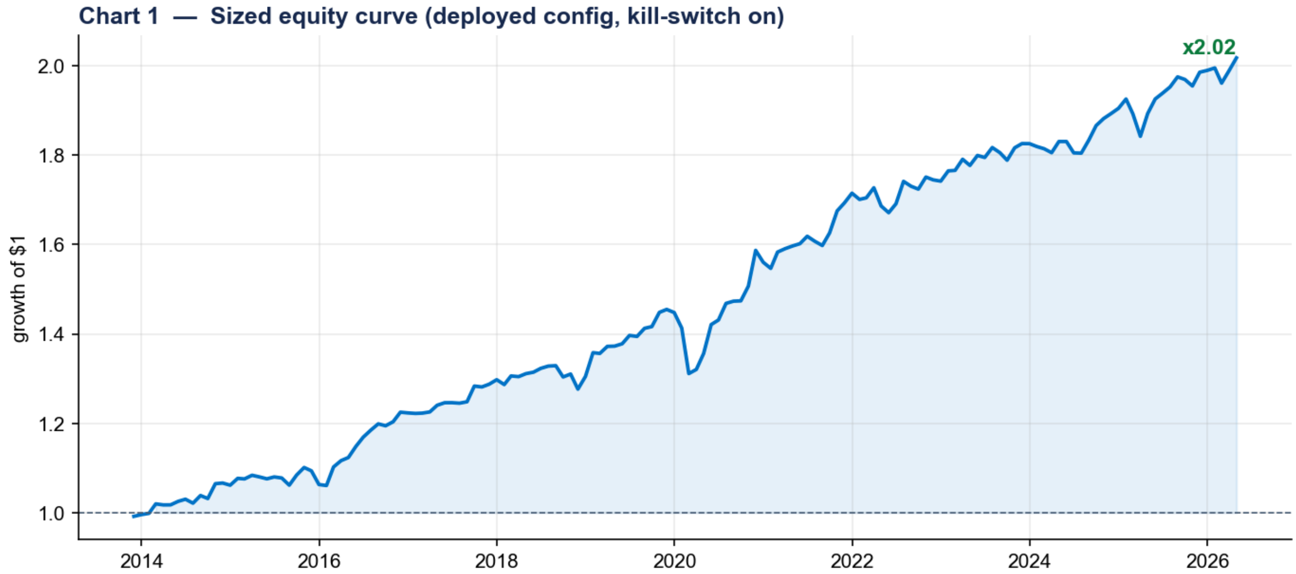


Chart 1 — The sized equity curve (deployed configuration). Growth of $1 from the 2013-11 warm-up boundary through 2026.

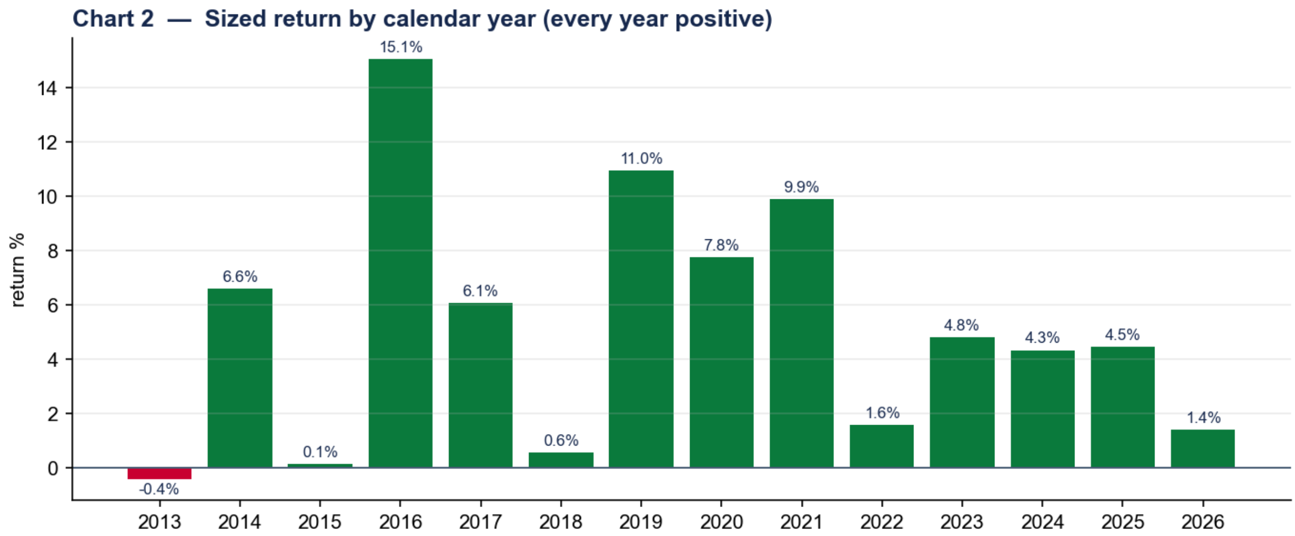


Chart 2 — Return by calendar year. Every year is positive — the breadth (hundreds of small, independent fades per month) is the source of the smoothness.

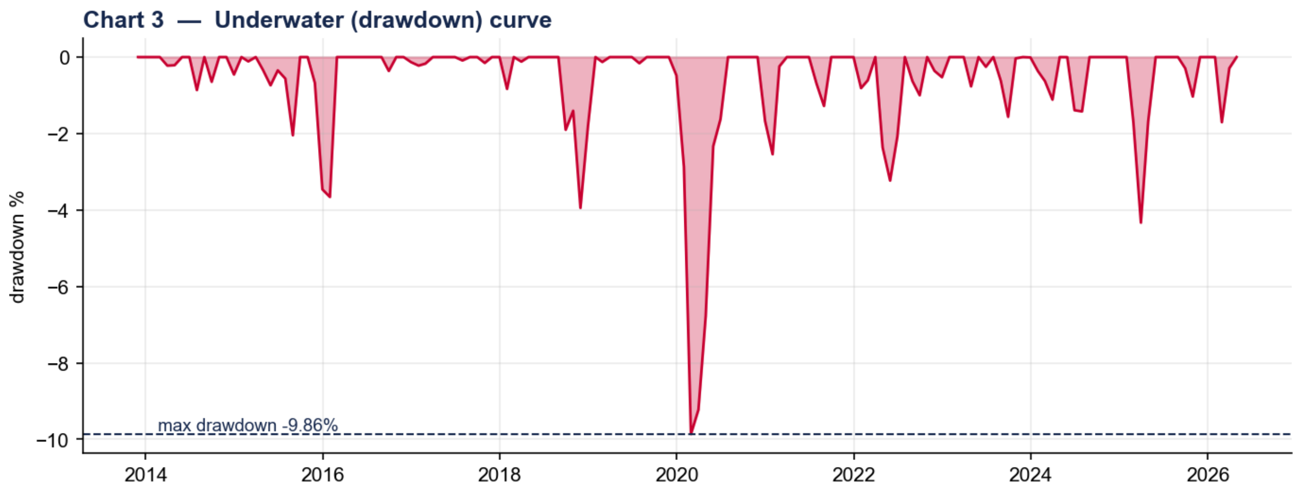


Chart 3 — The underwater curve. Worst drawdown –9.86%; the kill-switch's –15% trigger is never reached at deployed sizing.

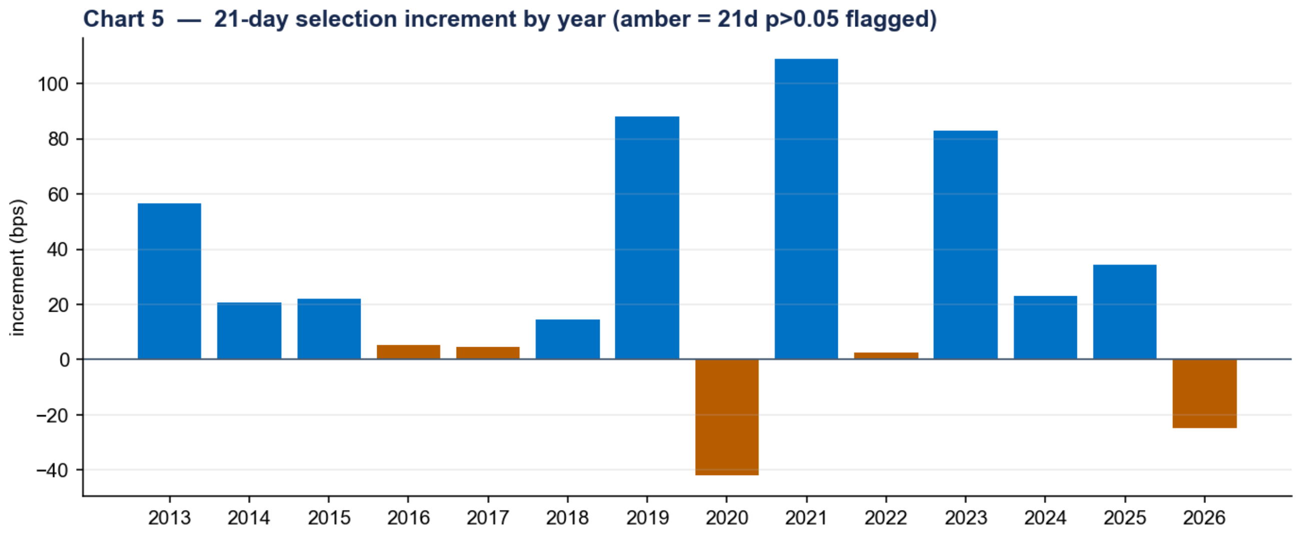


Chart 5 — 21-day selection increment by year. The edge is positive in most years; 2016–2017 are honestly flagged (21-day p > 0.05) — the system reports its weak years rather than hiding them.

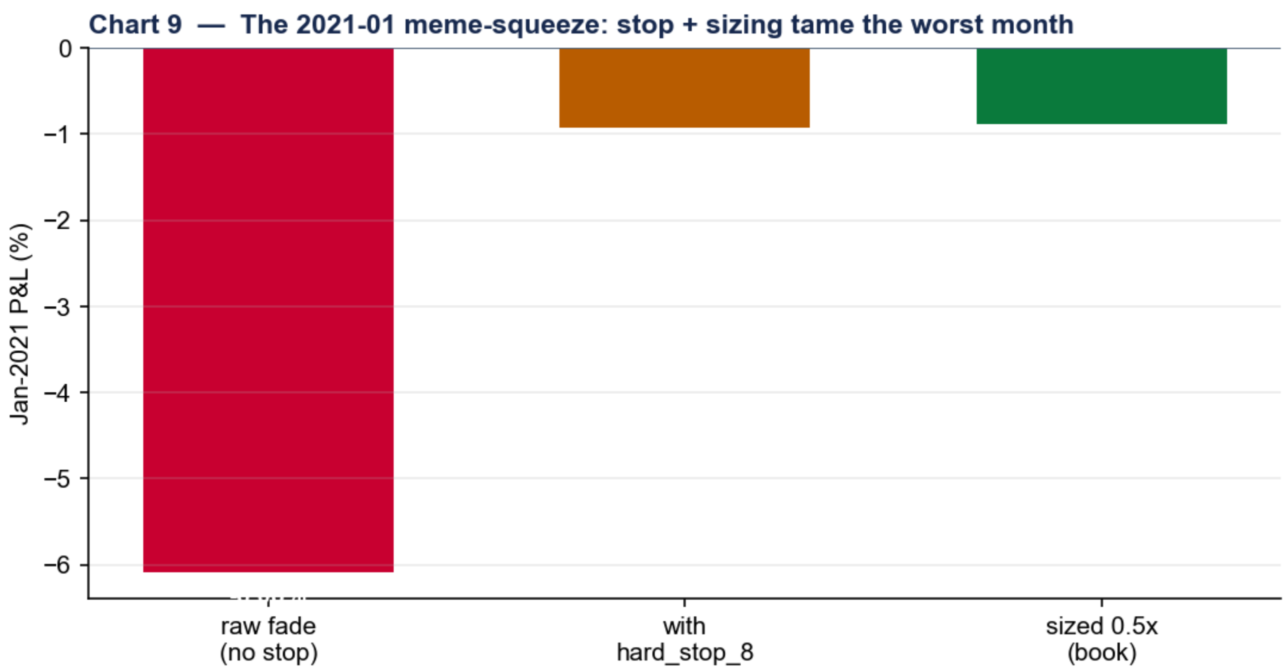


Chart 9 — The 2021-01 meme-squeeze, the worst case. The raw fade was run over; `hard_stop_8` *turns the month around (+516 bps vs no stop) and 0.5× sizing brings the booked month to –0.88%.*

> **The honest read.** The edge is real and robust — it beats a date- and direction-matched random pool by 18 bps over 21 days with z 8.87 across ~455k fires and 13 years, survives a multiple-testing-aware Deflated Sharpe, and holds across most regimes — but it is **economically modest** (a ~1.1 Sharpe) with one known weakness: a sharp V-shaped recovery runs over the fade (2020). That weakness is exactly why the optimisation layer is constrained rather than free-running, and why the next milestone is a paper track record before any leverage increase or live transition.

27b. Limitations, failure modes & roadmap

An honest system documents where it is weak. These are stated plainly so they are managed, not discovered.

Limitation / failure mode	Consequence	Mitigation / status
V-recovery regime risk	A sharp, broad V-shaped rebound (e.g. 2020) runs over the short side of the fade — the worst structural exposure.	Earnings/short-trend filters + `hard_stop_8` + the kill-switch bound the damage; not eliminated.
Economically modest edge	~1.1 Sharpe — real but not spectacular; sensitive to cost assumptions.	Costed at 20 bps; cost-survival breakeven tracked; deployed at 0.5×.
Short-side capacity	~10% of intended shorts are non-shortable on a given day (dropped, not forced).	Shortability gate skips + logs; net exposure cap absorbs the imbalance.
Single live data vendor	The signal depends on the ORATS surface; a feed outage stalls new fires.	Staleness guard refuses signals > 2 bdays old; manage loop still runs exits.
Lenient research DSR ceiling	The 0.95 DSR p-value gate is deliberately permissive for exploration.	Code comments flag tightening toward 0.05–0.10 for production; the optimiser already uses 0.95 as a *floor* for promotion.
No paper track record yet	Live OOS performance is unproven; backtest ≠ live.	Phase-5 (live) is gated on an accrued paper record (≥ 60 sessions, live Sharpe ≥ 0.5× backtest).
Sector concentration unmodelled	GICS sector caps are a parameterised hook, not yet active (no sector map in panel).	Per-name 5% + net 0.5 caps limit single-name and directional risk meanwhile.

Roadmap. (1) Accrue the paper track record on the live ORATS feed — the immediate next step. (2) Then the live transition (Phase 5): identical code, `paper=False` + live keys. (3) Longer-horizon: tighten the production DSR ceiling, add the sector map to activate sector caps, and graduate the Phase-2 `features_custom.*` modules onto the sandbox allowlist behind an import-safety audit.

27c. Incubating your own thesis — a practical guide

The pipeline is generic: any prose trading idea can travel it, not just the skew fade. This is how an analyst incubates a new thesis and what to expect at each gate.

What you write

A short `thesis.md` in plain English. The Hypothesis-Refiner does the formalising — you do not write JSON or code. A good thesis states: the **signal** (what you observe), the **action** (what you trade and which direction), the **horizon** (how long you hold), and — most usefully — **when you expect it to work and when to break**.

Strong thesis (passes refining)	Weak thesis (gets sent back)
“When 30-delta put skew spikes into an earnings drift, buy the underlying for ~2 weeks; works in calm tapes, breaks in a vol shock.”	“Oversold stocks bounce.” (no parameters, no regime, no horizon)

What the system needs & refuses

- **Data must be on the available list.** If your idea needs a feed the system can’t fetch (or a subscription it lacks), the refiner flags it and the run pauses rather than fabricating data.
- **The traded instrument sets `market_type`.** Using options data to trade the stock is `equities` — this changes which Pre-Critic template screens you.
- **It will try to kill your idea early.** The Pre-Critic is adversarial by design (“when in doubt, KILL”). A survivable concern becomes a warning; a fatal one (e.g. mid-price option fills) stops you before any compute is spent.
- **You cannot tune your way to a pass.** Every variant you try raises the Deflated-Sharpe bar via the cumulative trial count — quietly re-running with new parameters makes the next “win” harder to clear, on purpose.

How to read the verdict

1. **Did it beat the random pool?** The selection increment and z (stage 6) are the first real signal — a positive Sharpe alone is not enough.
2. **Did it survive the Deflated Sharpe?** Check `dsr_pvalue` against the trial count — a great Sharpe at 200 trials may be noise.
3. **What did the Validator flag?** Warnings are acceptable and logged; a fatal needs a written, ≥ 20-character override that lives forever in the audit trail.
4. **How big did Risk allow?** The `size_recommendation` (default 0.5×) and the concerns list tell you how much the officer trusts the tails.

> **The contract you get back.** A single `decision.json` (✓ accepted / ✗ rejected / ■ accepted-with-override) plus a living HTML report you can open in a browser or print to PDF — every stage, every number, every override, on one page. If accepted, the strategy can be deployed to DRY-RUN paper trading with one command and graduated to real paper fills with an explicit `--submit`.

28. Glossary

Term	Meaning
Fade	Trade *against* a crowded extreme, expecting reversion (BULL corner → short; BEAR corner → long).
Fires frame	The canonical strategy output: one row per (ticker, date) signal with side, sign, forward returns, and stage flags.
Selection increment	How much the signal beats a same-day, same-direction random pick — the engine measures selection skill, not market drift.
Deflated Sharpe (DSR)	A Sharpe haircut for the number of trials tried, so data-mining a good-looking number doesn't pass.
Materialization parity	The stored panel side that the verdict actually scores must equal a fresh reference recompute — bit-for-bit.
Kill-switch	A hysteretic drawdown governor that de-risks the book to 0.3× at –15% and restores at –7%.
Idempotent client_order_id	A deterministic order key so re-running the daily loop never double-submits.
OPG	Market-on-open time-in-force, matching the backtest's next-open fill basis.

29. Constants & thresholds quick-reference

Domain	Constant	Value
Signal	hi / lo percentile, freshness, sigma_threshold	75 / 25, 3 bars, 1.0
Signal	divergence threshold (d3 vs d2)	± 0.20
Sign	BULL / BEAR	–1 (short) / +1 (long)
Screen	price floor, max |forward return|	$1, 5.0
Backtest	horizons, annualisation, cost	5 / 10 / 21, 252, 20 bps
Vs-random	bootstrap draws	2000
Gates	DSR p-value max, correlation max, PCA share max	0.95, 0.60, 0.50
Sizing	Kelly λ, correlation ρ	0.25, 0.30
Sizing caps	per-name, gross, net	5%, 1.0, 0.5
Kill-switch	enter / de-risk / exit / hard	–15% / ×0.3 / –7% / –25%
Risk	hard-reject score, default deploy	≥ 8, 0.5×
Exits	deployed policy	hard_stop 8% + Bollinger reversion band + 21D backstop

Optimiser	prior trials, prob-real bar, DD bound, edge-keep	12, 0.95, –0.25, 0.50
Paper	staleness, order type, default mode	≤ 2 bdays, OPG, DRY-RUN
Data	warm-up, panel start anchor	756 bdays, 2013-11-26
Parity	tickers, tolerance	AAPL/MSFT/XOM/JPM, 0

30. Audit-trail map

Every run leaves a complete, append-only paper trail under `theses/<id>/`, so any verdict can be reconstructed and any override audited.

Stage	Artifact
0–1	`thesis.md` → `refined.json` + `step_summaries/01_refiner.md`
2	`critique_pre.json`
3–4	`code/signal.py`, `results/{vs_random.json, returns.csv, positions.csv, net_return_panel.csv}`
5–6	`results/{metrics.json, dsr.json, walk_forward.json}`
7–9	`critique_post.json`, `gates_outcome.json`, `risk.json`
10–11	`memory.db` row, `reports/<strategy>/report.{json,html}`, `decision.json`
cross-cutting	`audit_log.jsonl`, `user_interactions.jsonl`, optimisation `reports/{mutation_agent,exit_agent_selection}.json`

— End of report —

Confidential working document · Skew-Consensus Validation System · Extended Companion to Project Record v2.2 · May 2026. Every numeric value above is sourced from the codebase on branch `claude/funny-elbakyan-1e8456`.